



Fig. 6: Servo attachment with valve



Fig. 7: Testing arrangement

for interfacing the GSM module and then the servo library. Next, define variables for the pin numbers of the connected components. After that, in the setup function, serial is started with the default baud rate of the SIM9000A module. After completing other setup tasks, the loop function is created to check the LPG gas sensor status.

If the sensor detects gas, the message sending function and servo function are activated. You can also add a buzzer or doorbell ring and other functions to the code that needs to be executed after detecting LPG gas leakage. Fig. 3 shows a source code snippet defining the pins.

To upload the source code, select the correct COM port and Arduino board type, and then upload the code. After that, connect the compo-

nents according to the circuit diagrams shown in Figs. 4 and 5.

Circuit and working

There are two circuits in this project. The first circuit is used with a buzzer and LED, as shown in Fig. 4, while the second circuit is used with a relay and relay driver IC

ULN2003, as shown in Fig. 5. The source code for both projects is the same.

In this proposed project, we have interfaced Arduino Uno with MQ2/MQ6 sensors, SIM900A GSM module, servo motor, buzzer, and LED. The interfacing is done in such a way that whenever there is a gas leak due to any fault, the MQ2/MQ6 gas sensor detects the leakage and sends a signal to Arduino Uno in the following manner:

1. To drive the servo motor attached to the gas cylinder regulator, so that it immediately switches off the regulator automatically.

2. To drive the SIM900A GSM module to send an alert message to a mobile number.

3. To switch on the bell/siren as an alert.

Construction and testing

Assemble the circuit on a bread-board or general-purpose PCB based on the circuit shown in Fig. 4 or Fig. 5. But do not forget to upload the source code into the Arduino Uno. It would be better if you refer to Fig. 6 and Fig. 7 before assembling to get an idea of it. Fig. 6 shows how the servo motor is attached to the valve.

For testing, refer to Fig. 7. After uploading and assembling the components, power on the circuit, including Arduino and other components, using a 5V adaptor. When gas leakage is detected, it automatically turns off the valve using the servo and then sends you a message while giving an alert using the buzzer and LED (buzzer sounds and LED glows).

EFY note. Here we have used a buzzer and LED as an alert indicator, as shown in Fig. 4. But in Fig. 5, we have used the ULN2003 IC and an electric bell.

This project is very important for gas leakage safety purposes and definitely for the safety of life and property.

Advantages

1. Detection of gas leakage
2. Automatic tripping of the regulator knob to the off position, which can be reset by the user.
3. Automatic indication and message facility. **EFY**

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EFY Note
The source code
of this project
is available

for free download at
www.electronicsforu.com